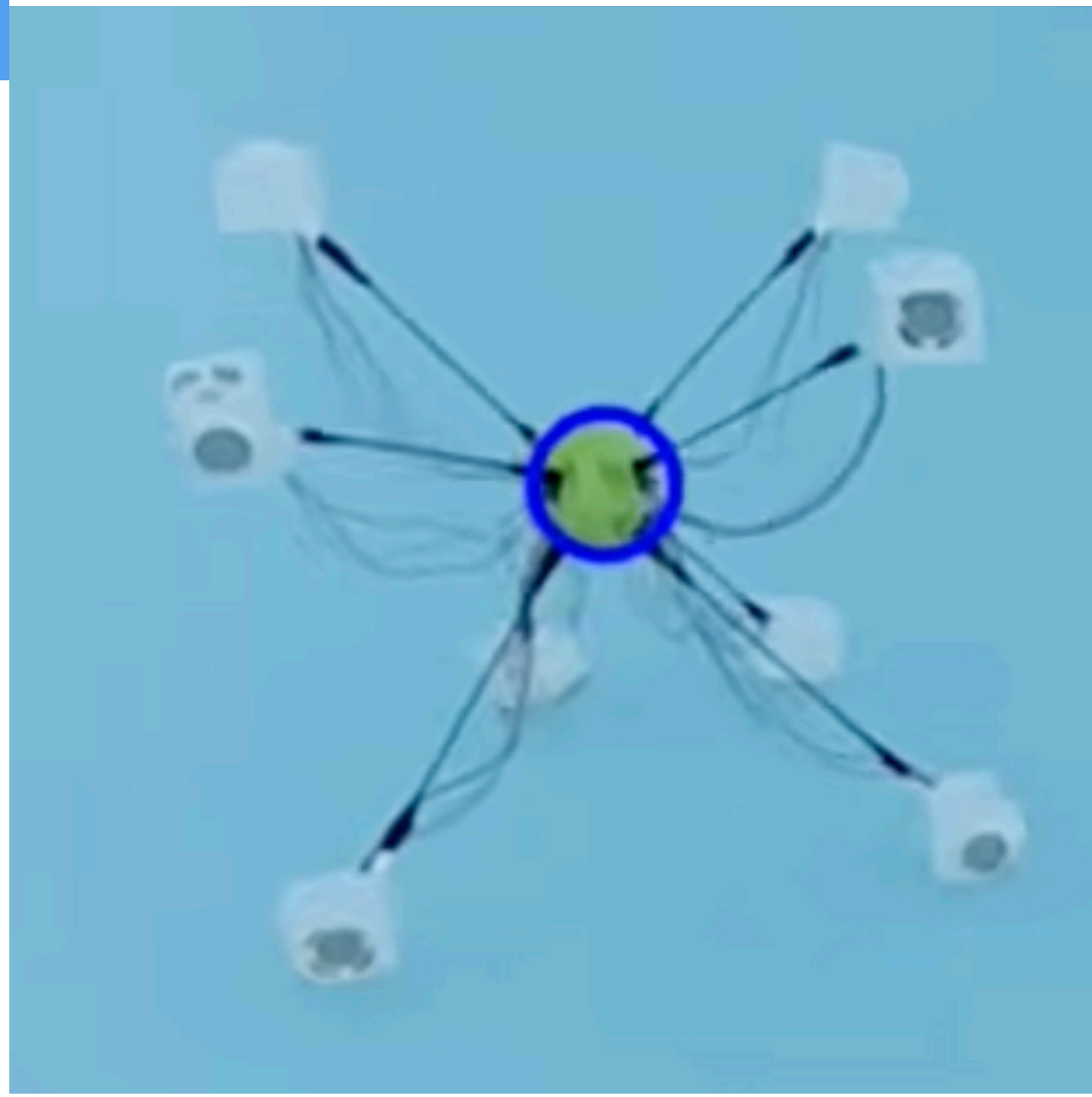
Workshop 2: Slides on the go !

Team Casuals

Hasith Perera, Justin Hsu, David Niemczura, and Jaylin Washington

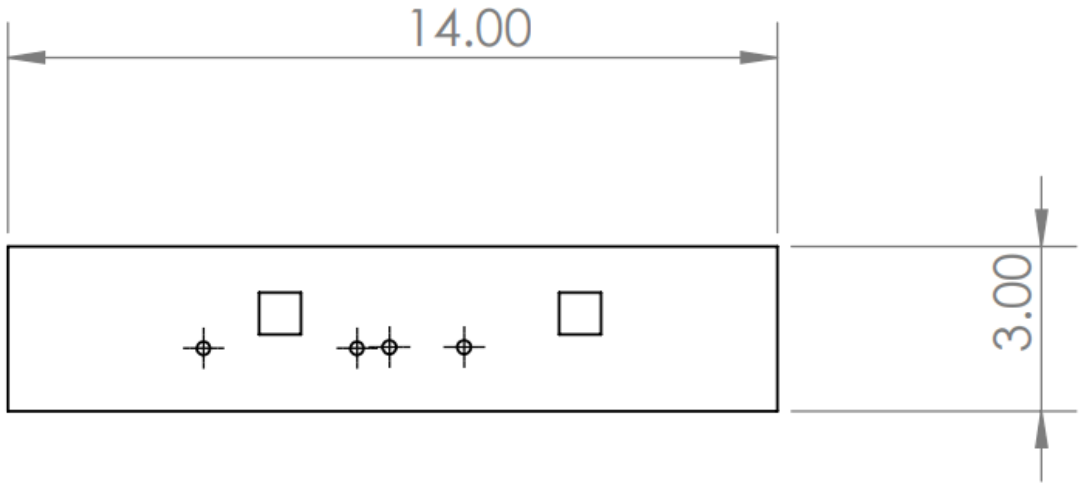
SMILE - Alternate design

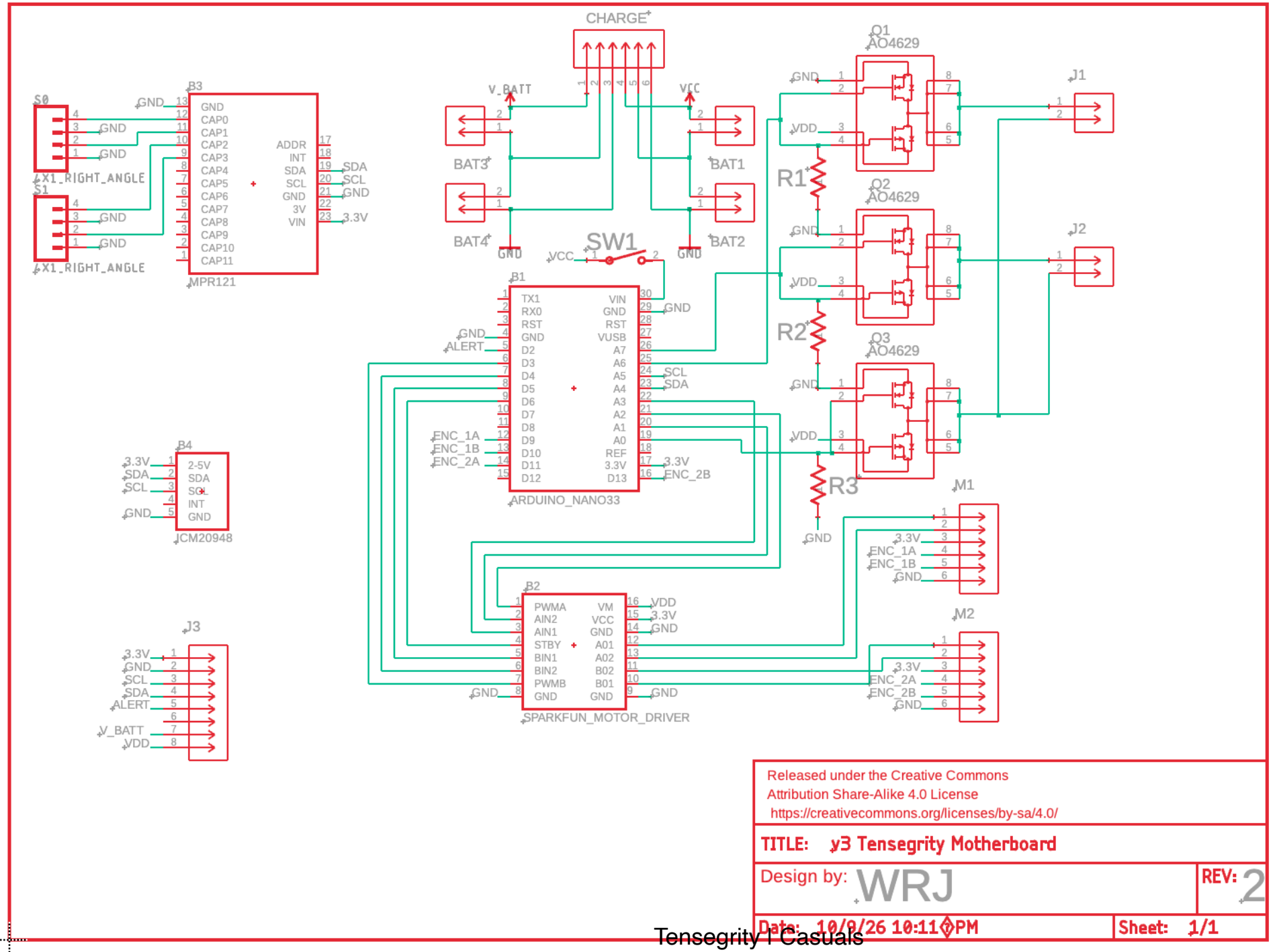
- Rejected due to design timeline constraints
- Would require custom Hardware and Electronic design



Yale Desing

- This is the Yale control board; the COTS board has everything you need and more, but the physical dimensions are not feasible
- We are leaning toward a custom PCB order with a minimum design
  - Remove force feedback, but maintain the form factor
  - Add the current module




CHARGE
V_BATT
VCC
BAT3
BAT1
BAT4
BAT2
GND
SW1
Q1
AO4629
Q2
AO4629
Q3
AO4629
R1
R2
R3
J1
J2
J3
M1
M2
B1
ARDUINO_NANO33
B2
SPARKFUN_MOTOR_DRIVER
B3
MPR121
B4
JCM20948
S0
S1
X1_RIGHT_ANGLE
X1_RIGHT_ANGLE
Released under the Creative Commons
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https://creativecommons.org/licenses/by-sa/4.0/
TITLE: y3 Tensegrity Motherboard
Design by: WRJ
REV: 2
Date: 10/9/26 10:11 PM
Sheet: 1/1

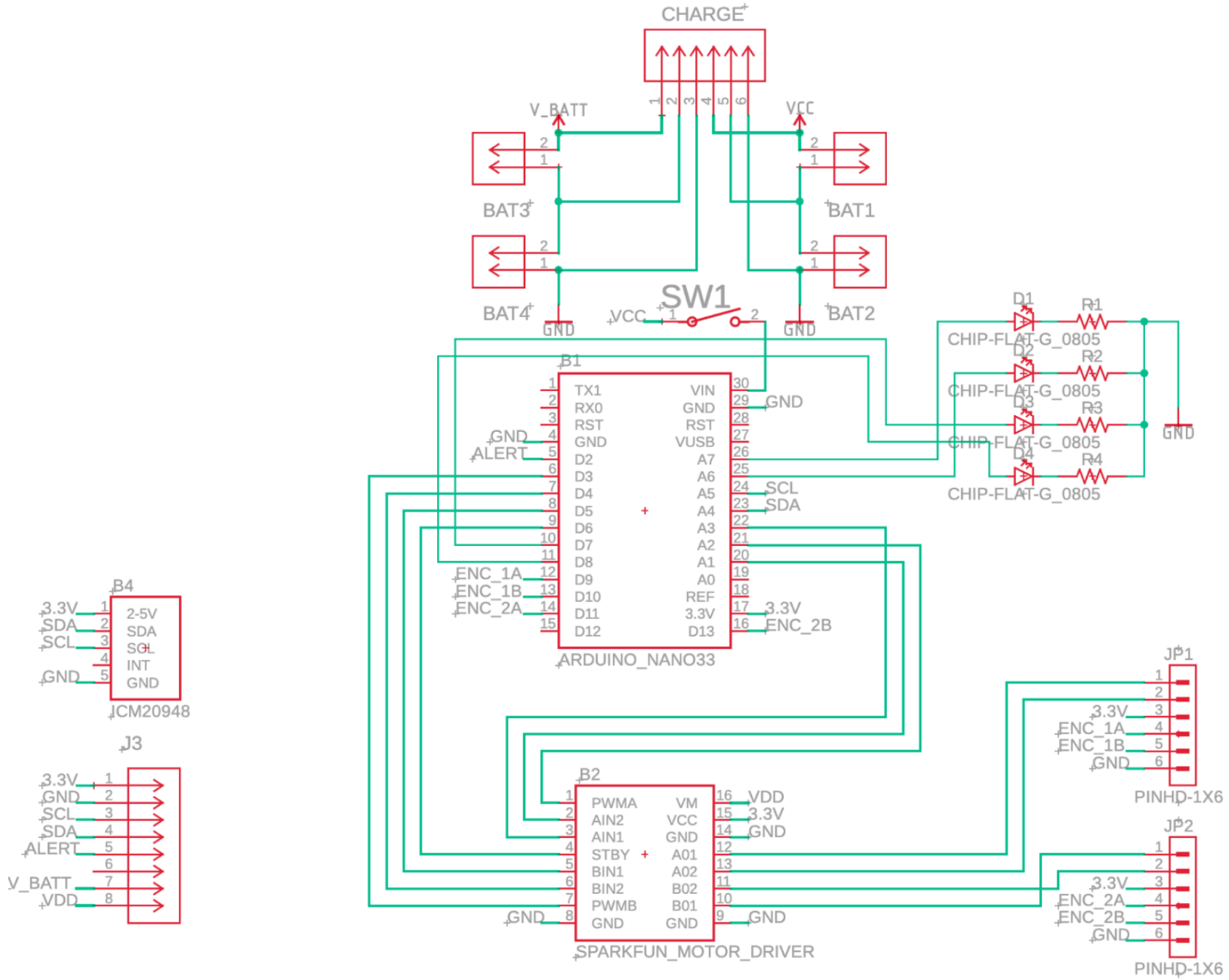


Figure 5: Modified PCB design after removing unwanted components

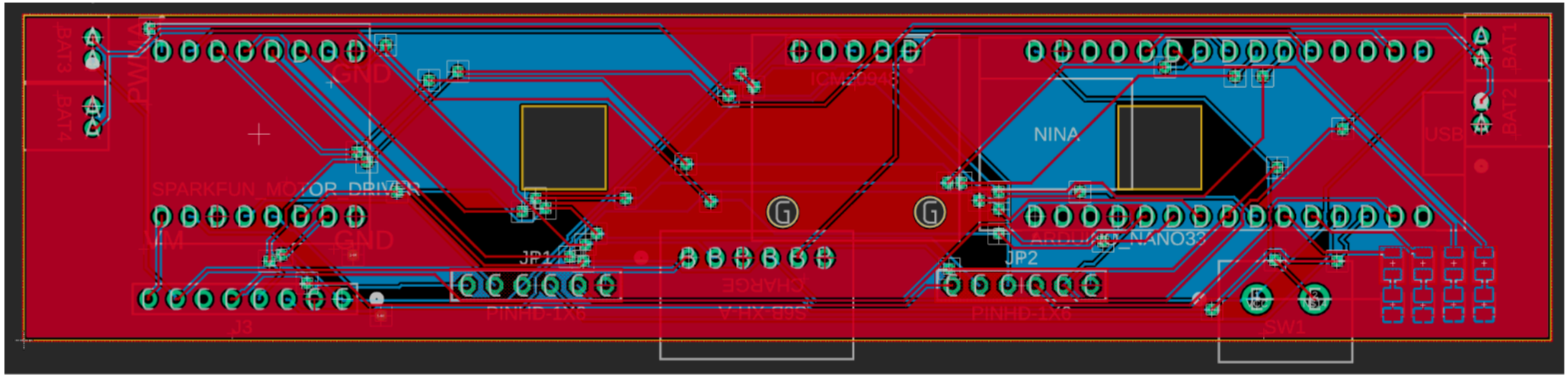


Figure 6: PCB design maintaining the original dimensions and all physical constraints

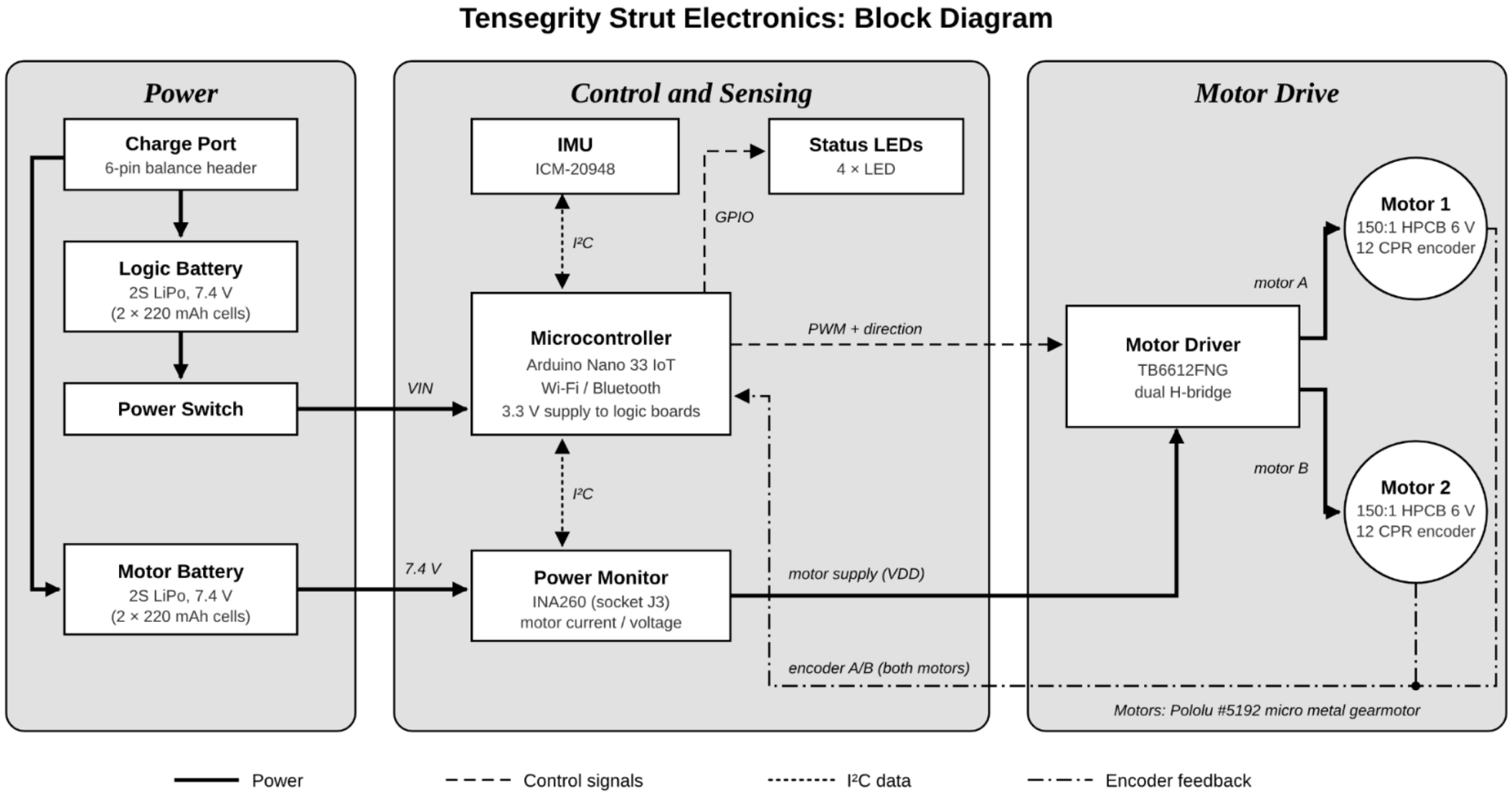


Figure 7: Block diagram of one strut's electronics. All three struts carry an identical copy.

Initial proposed time line

Description	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Proposal and BOM												
PCB design												
Electronics testing												
Mechanical building												
Make 2 copies												
Stage 1 control system												
Drop testing												
Movement												
Stage 2 control system (ROS)												

Note: Green boxes are tasks that are already being completed; blue indicates tasks that are pending in the future

Workshop 3: Update

Team Casuals

Hasith Perera, Justin Hsu, David Niemczura, and Jaylin Washington

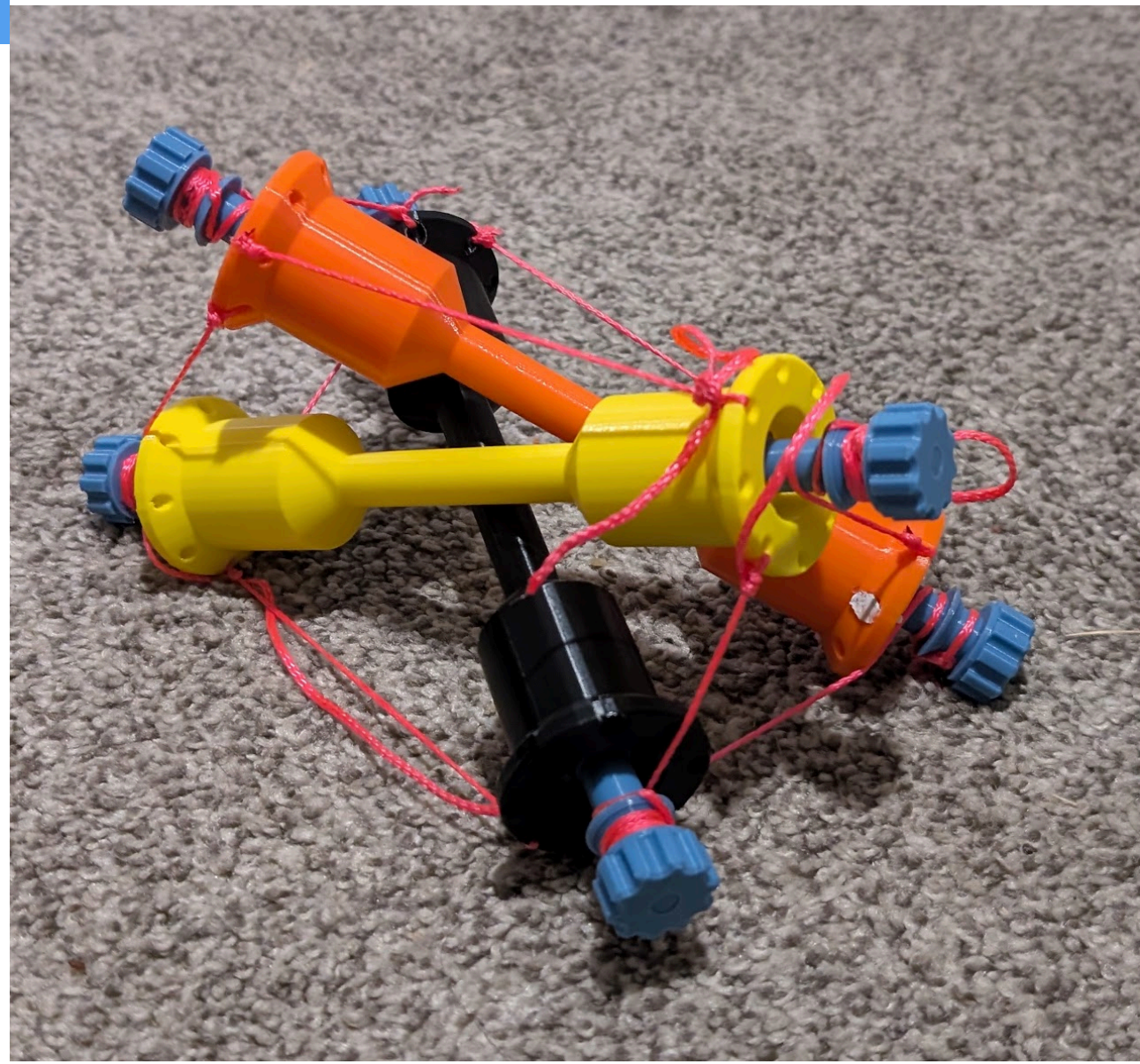
Updated and current time line

- Additional constrains and resubmission of the BOM added a week of delay
- Item review and delivery/pickup again added ~ 1.5 weeks
- Lab space/soldering access has not been provided yet

	PCB Order	delay due to BOM	delay - pickup	Hasith -work travel						Hasith - travel Dec 11		
	Oct 12	Oct 19	Oct 26	Nov 2	Nov 9	Nov 16	Nov 23	Nov 30	Dec 7	Dec 14	Dec 21	Dec 28
Description	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Proposal and BOM												
PCB design + delivery												
Component order + delivery												
Electronics testing												
Mechanical building												
Make 2 copies												
Stage 1 control system												
Drop testing												
Movement												
Stage 2 control system (ROS)												

Progress

- **Justin and Hasith** put together a model to see how this design will work and maintain stability
  - it works on a ratchet mechanism
- We found that the fixed string length is a constraint we have to tune. (This is not specified in any of the public documents)



Current issues we are thinking about

- The controller Yale design is using is a **Nano 33 IoT**
  - This is a single core
  - they are not using interupts in there code
  - using UDP packets to get data in and out of the 3 structs
  - no close loop controllers are running in the struts
    - Would putting 4 encoder signals break the speed required for UDP data
  - Should we upgrade to a dual core processor
    - can be done